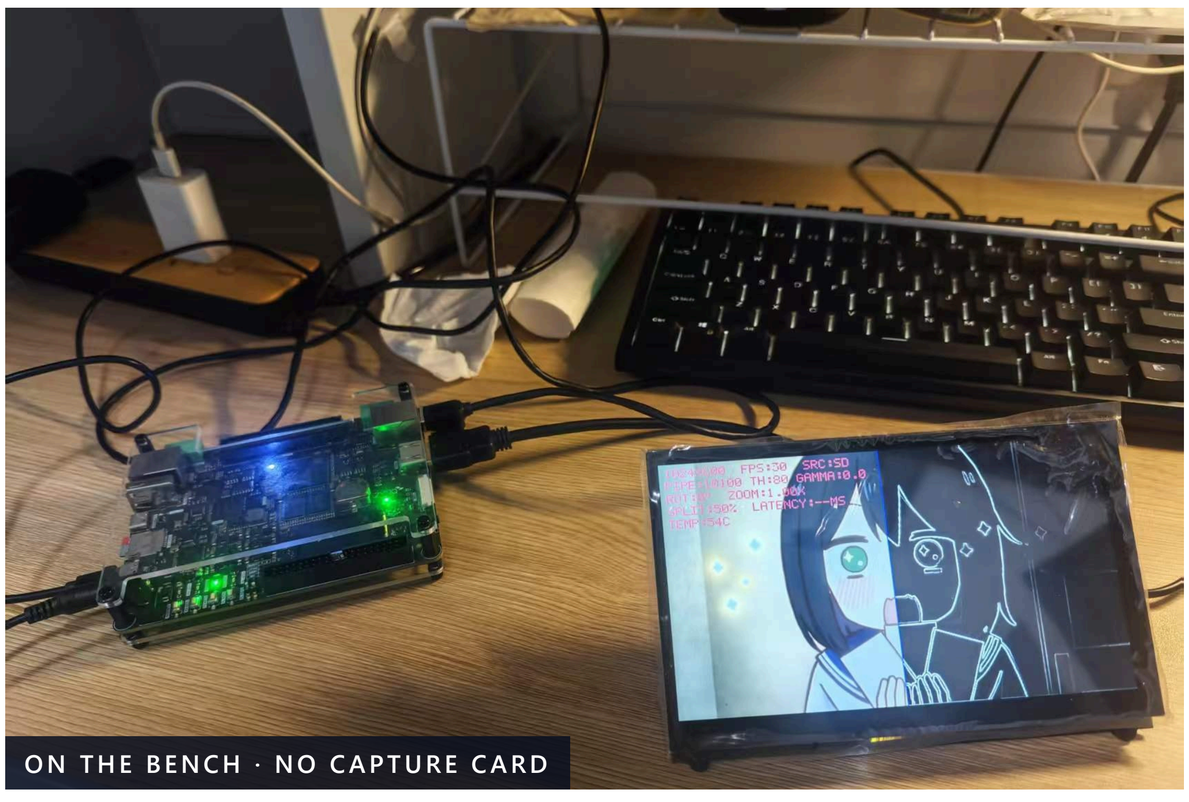
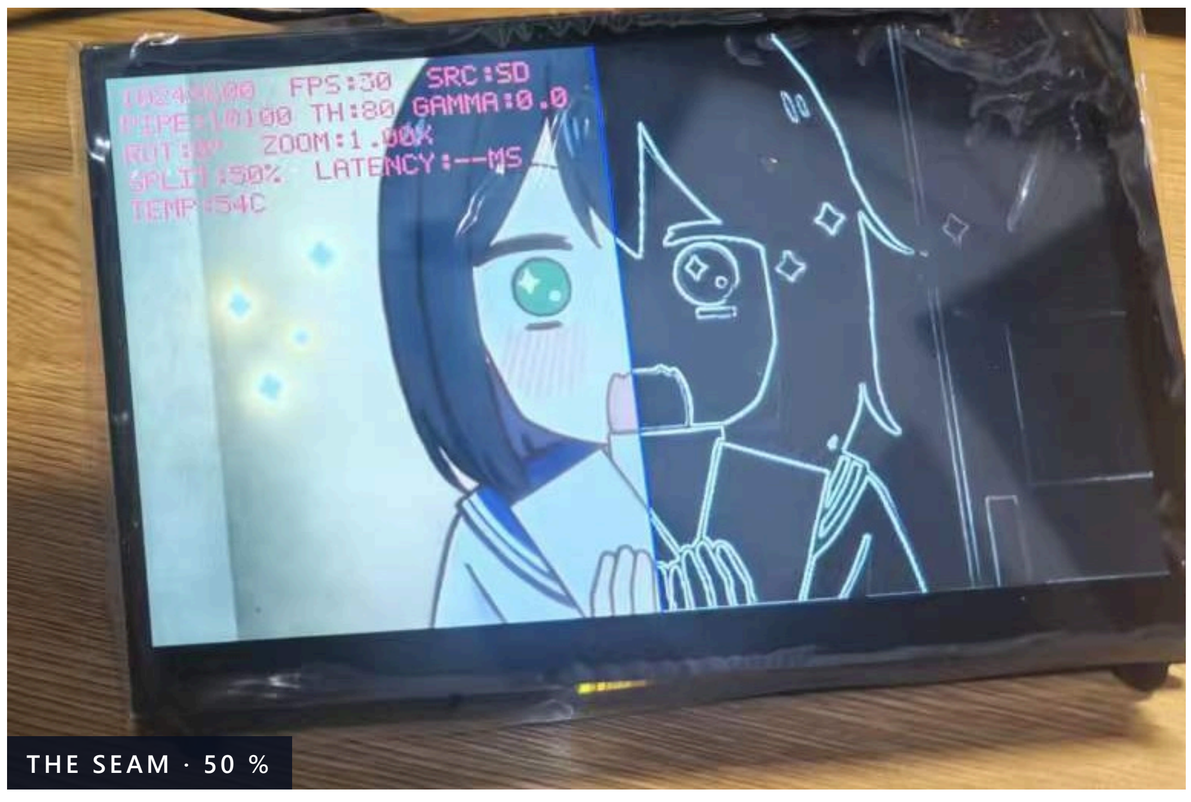
ONE FRAME. TWO PICTURES.

One read address, one clock cycle, one scanline — **the panel itself becomes the measuring instrument.**

Device **xc7z020clg484-2** Tools **Vivado / Vitis 2025.2.1** RTL **80 files · 11,768 lines** Firmware **2,544 lines bare-metal C** Build on panel **r132 · bit md5 cb907d3e4898**





Left of the seam the panel shows the **unprocessed** frame; right of it the **processed** one — same frame, same instant, one read address. The seam position is a runtime parameter (`split 0…100`), so a geometry error shows up as a step or a black wedge **on the panel**, not as a log line. OSD reads `SRC:SD · FPS:30 · SPLIT:50% · TEMP:54C` , and the same cells are read back over AXI GPIO.

165
pixel criteria · 0 FAIL
one top bench walks the whole design, per column, per clock

29.99
fps · 0 dropped
300 s / 9,000 frames, counted in hardware on the card

116.7
fps headroom
unthrottled 120 fps ingress, 960 frames, no lost word — a bound, not a maximum

+0.687
ns WNS · 0 failing
post-route, 0 of 51,027 setup and hold endpoints failing, TNS 0

68.6
% of BRAM tiles
96 / 140 tiles · LUT 26.3 % · registers 7.6 % · DSP 8.6 %

01 WHERE A PIXEL TRAVELS

Ingress
RGMII → GMII in the PL. IDDR + IDELAY, frame reassembly, port filter; ARP and ICMP answered in RTL.

Storage
DDR ping-pong. Three banks, BRAM row rings, a single-owner arbiter across ETH / SD / test card.

Geometry
Zoom · rotate · bilinear. Eight fixed-point ratios 0.25–2.00× plus auto-fit; 0–359° stays inside the panel.

Effects
Six stages, nine switches. One 9-bit control word, fixed **15-cycle** latency for any combination.

Display
A/B seam → OSD → TMDS. 1024×600 @ 59.5 Hz off a 50 MHz pixel clock, ×2 serialised.

02 WHY THE PICTURE CAN BE TRUSTED

INVARIANT

One coordinate operator, fixed point
`zfit` solves the fit factor in 3 DSP48 blocks; `zmap` maps every pixel through 8 DSP48 blocks, rotation running **inverse-mapped** with two linear interpolations. Eight ratios, auto-fit on top, so a rotated frame **always stays inside the panel** — checked by a two-sided bench, not by an eye.

INVARIANT

Latency is a constant
Six pipeline stages, each window stage keeping its own column tag, so total latency is **15 cycles and content offset −4 rows for every combination**. The fabric exposes one number, `u_pipe.LATENCY` ; the read scheduler never re-derives it. Across the frame boundary the read-side wrap is keyed to the ring's write beat, so the previous frame's pixels are fixed, not stale. Any drift fails a bench test.

INVARIANT

Every claim can go red
A criterion without a way to fail is not evidence. **83 benches** carry a mutation control — flip one bit, delete one line, and the bench must go red — and a **24-item gate** runs before release. A zero-hit scan is reported as a blind ruler, never as a pass.

24-item gate, 0 red
one command, last line is the verdict. Two runs inside this round went red on stale credentials; both were fixed in the checkers, not by loosening a criterion.

4 optimisations declined
forced replication of the 239-fanout net, a Pblock over the FIFO BRAMs, BRAM-for-setup, a ±0.5 ns input window — published with their readings, including the ones that won one cell and lost four.

Measured, estimated, or not claimed
Junction temperature on the running board is XADC 61.2–61.7 °C (r113 build); the r132 figure 52.6 °C is a tool estimate at confidence Low. QSPI image written and verified, but no cold-boot capture on file, so no auto-start claim.

```
vivado -mode batch -source build/build.tcl · project → synth → impl → bitstream, about 20 min
bash build/gates.sh · 24 pre-release items; the last line is the verdict
bash build/sim/run_one.sh tb_v98_top_seam · 165 per-pixel criteria, elapsed 03:21:55
bash build/board_verify.sh --geom --battery · geometry check + 105 serial commands against the panel
```

Readings come from one build: **bit md5 `cb907d3e4898`**, reports archived with the submission.